

C: Pointers and Arrays

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What we've learnt in last lecture

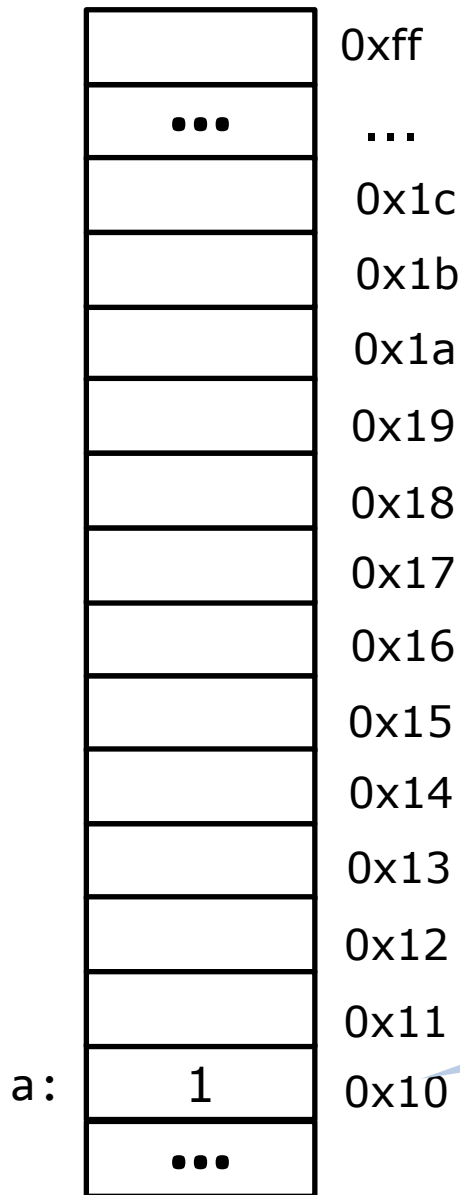
- C program's basic organization
 - *.c vs. *.h files
 - Compilation workflow and make
- Bitwise operators, &, |, ~, ^, >>, <<
 - >> (arithmetic vs. logical)
- Control flow
 - goto is general, but results in spaghetti code

Pointers

Pointer is a memory address

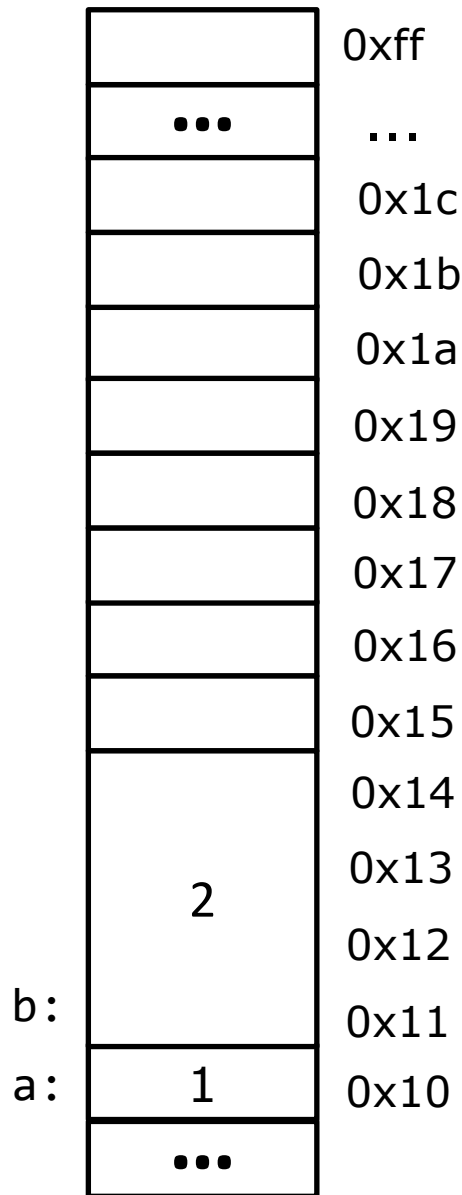
Pointer

```
char a = 1;
```



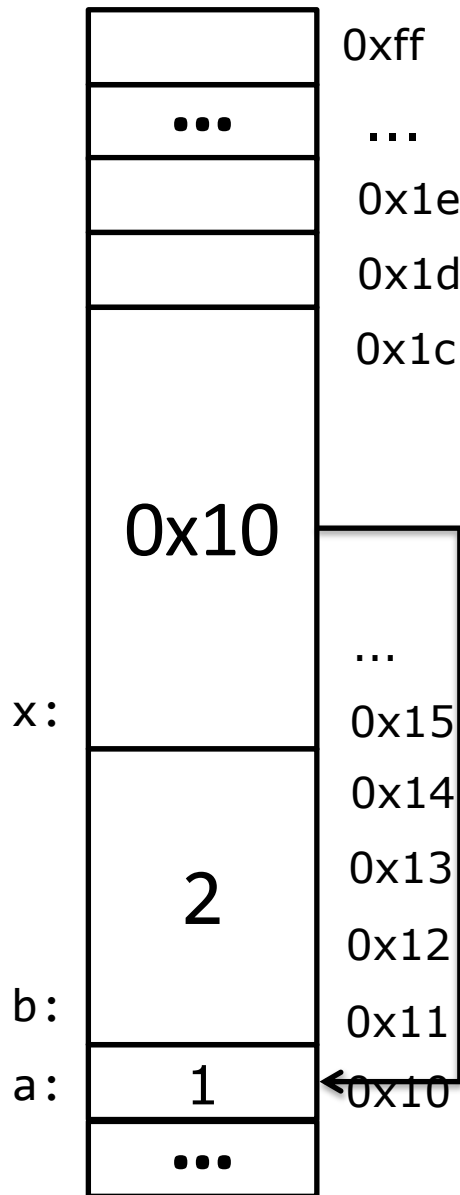
Addresses are 8-byte long on 64-bit machine; for brevity, I omit leading 0s.

Pointer



```
char a = 1;  
int b = 2;
```

Pointer



```
char a = 1;
```

```
int b = 2;
```

```
char *x;
```

```
x = &a;
```

Same as: `char* x;`
You pronounce typename
from right to left

& gives address
of variable

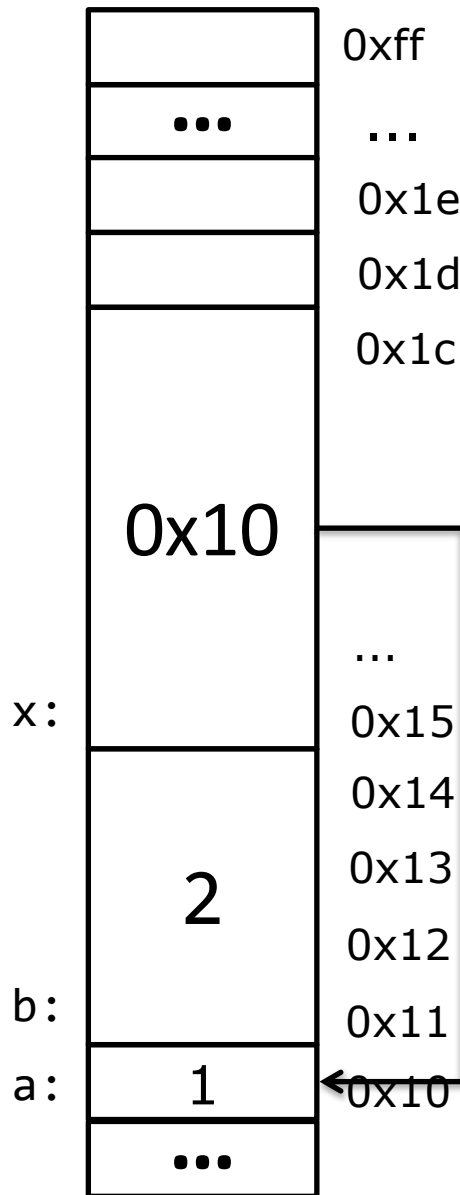
Can be combined as:
`char *x = &a;`

what happens if I write
`char x = &a;`

type mismatch!

Pointer

```
char a = 1;  
int b = 2;  
char *x = &a;  
printf("x=%p\n", x);
```



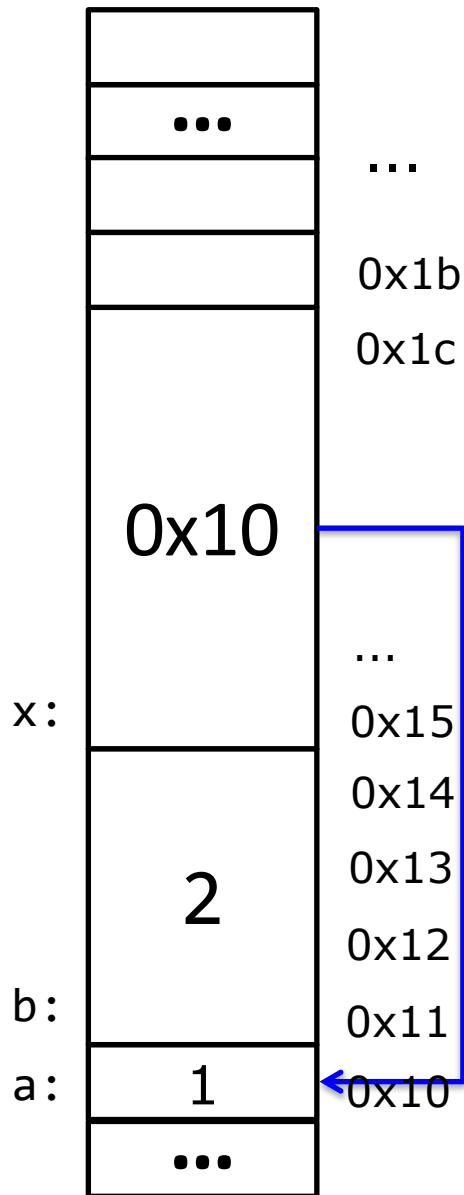
You can print the value of a pointer variable with %p (leading zeros are not printed)
0x10



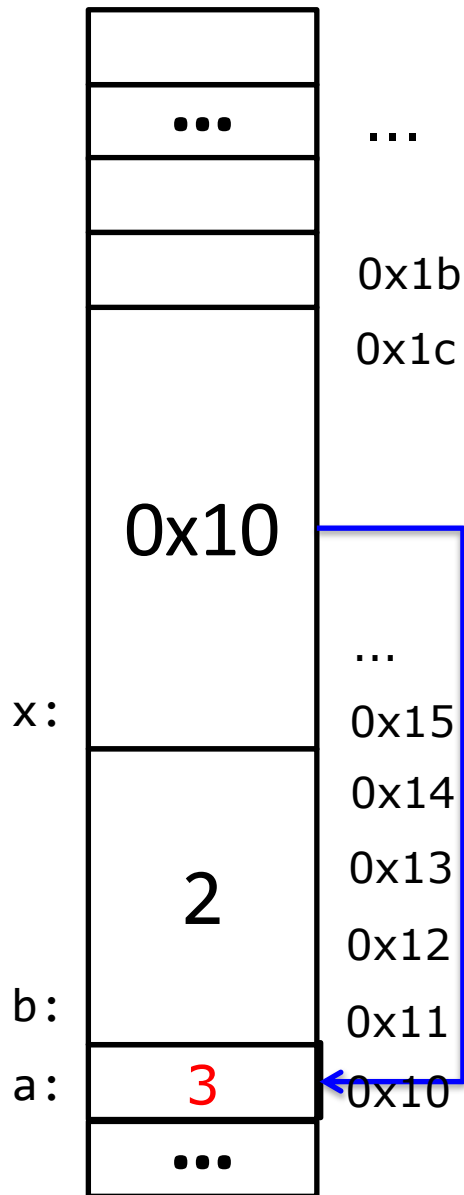
Size of pointer on a 64-bit machine?
8 bytes

Pointer

```
char a = 1;  
int b = 2;  
char *x = &a;
```



Pointer



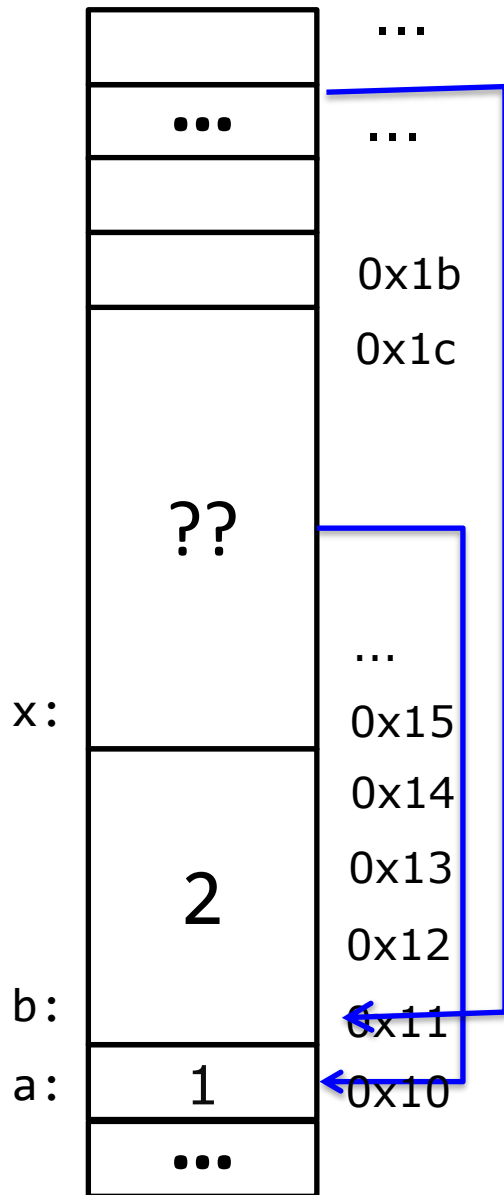
```
char a = 1;  
int b = 2;  
char *x = &a;
```

```
*x = 3;
```

* operator dereferences a pointer, not to be confused with the * in (char *) which is part of typename

Value of variable a after this statement?

Pointer



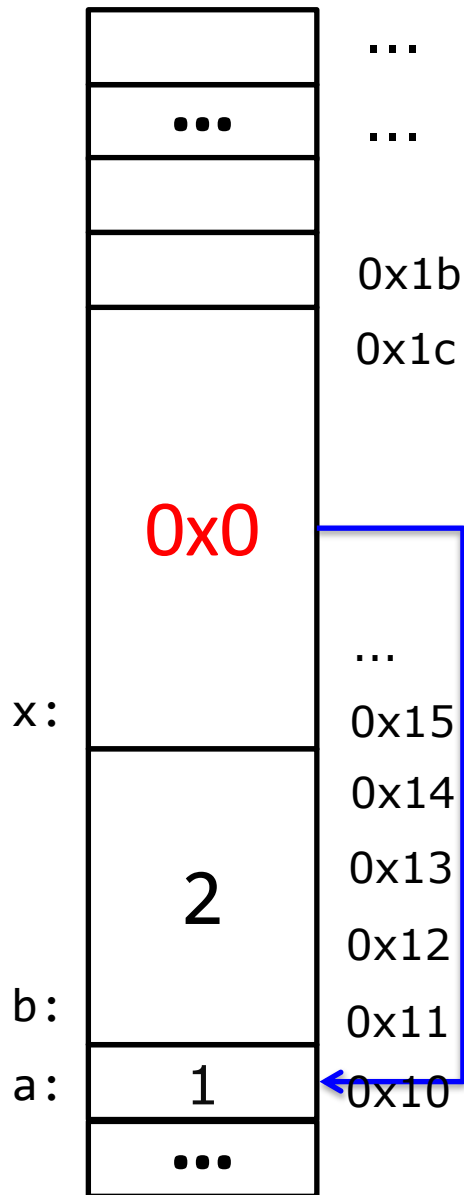
```
char a = 1;  
int b = 2;  
char *x = &a;
```

what if x is uninitialized?

```
*x = 3;
```

Dereferencing an arbitrary address value may result in "Segmentation fault" or a random memory write

Pointer



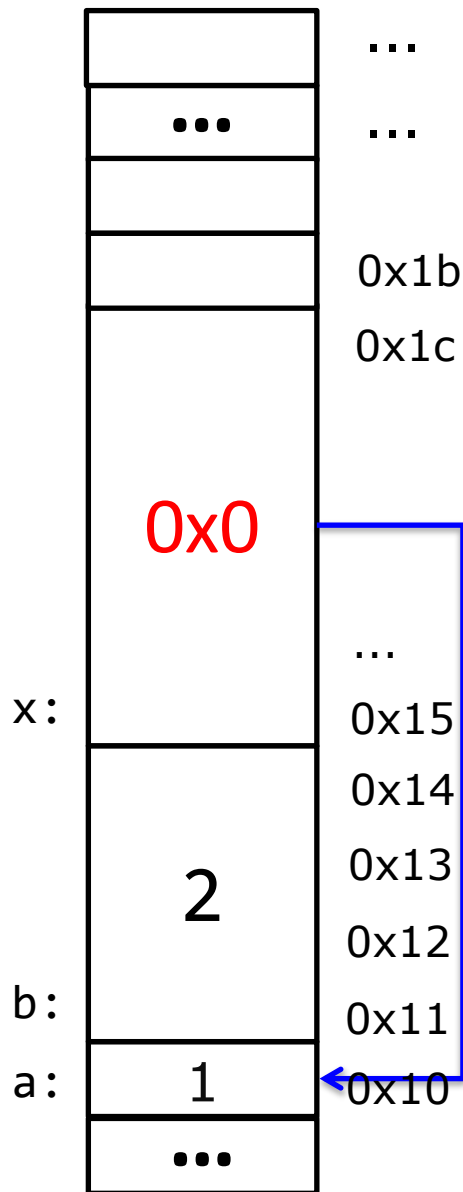
```
char a = 1;  
int b = 2;  
char *x = NULL;
```

Always initialize
pointers!

```
*x = 3;
```

Dereferencing NULL pointer
definitely results in
“Segmentation fault”

Pointer



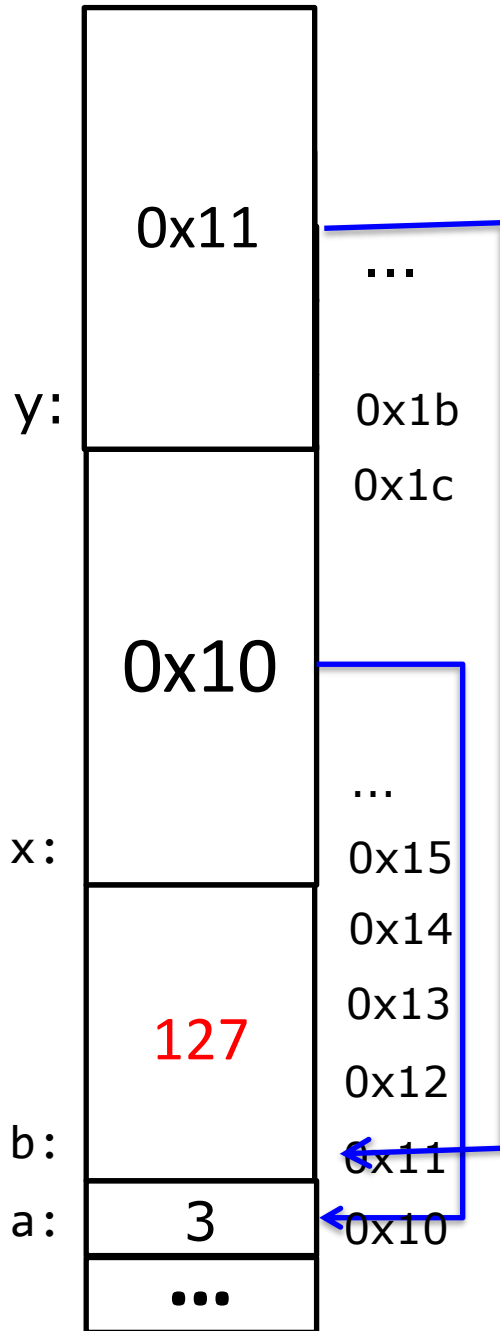
```
char a = 1;
int b = 2;
char *x = NULL;
```

```
*x = 3;
```

```
(gdb) r
Starting program: /oldhome/jinyang/a.out

Program received signal SIGSEGV, Segmentation fault.
0x00000000004005ef in main () at foo.c:16
16          *x = 3;
(gdb) p x
$1 = 0x0
(gdb)
```

Pointer has different types



```
char a = 1;  
int b = 2;  
char *x = &a;  
  
*x = 3;
```

```
int *y = &b;  
  
*y = 127;
```

what if I write
char *y = &b;

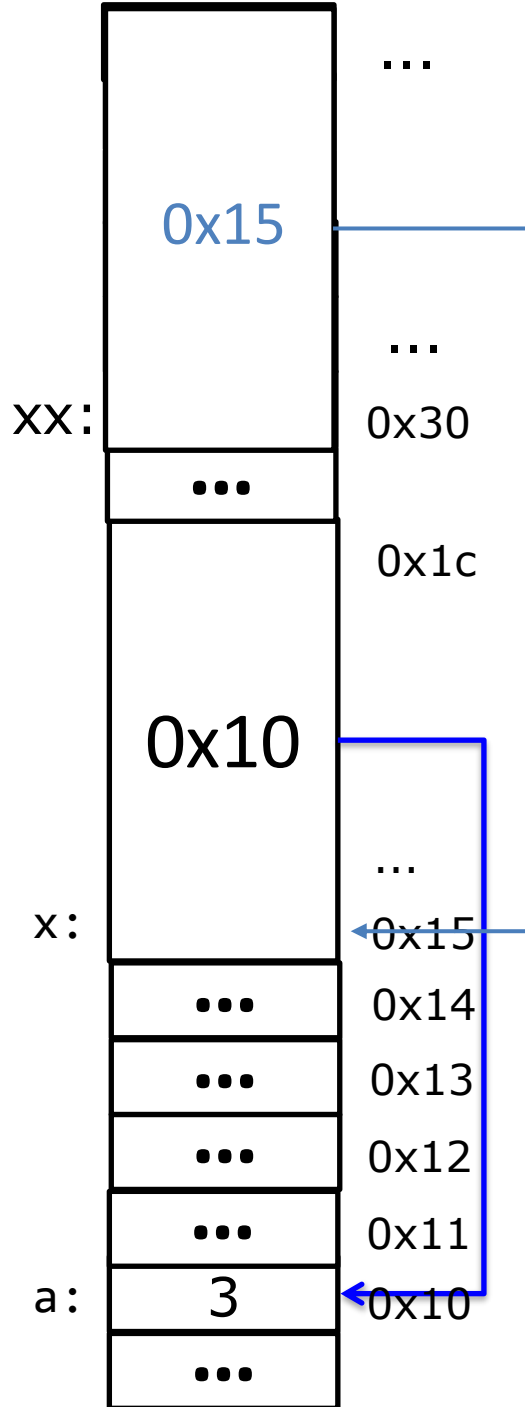


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Double Pointer



```
char a = 1;
int b = 2;
char *x = &a;
*x = 3;
```

```
char **xx = &x;
```

char **xx is the
same as char** xx;

Same as:
char **xx;
xx = &x;

what if I write
char *xx = &x;

```
printf("xx=%p *xx=%p **xx=%d\n", xx, *xx, **xx);
```

Common confusions on *

* has two meanings!!

1. part of a pointer type name, e.g. `char *`, `char **`, `int *`
2. the deference operator.

```
char a = 1;  
char *p = &a;  
*p = 2;
```

```
char *b, *c;  
char **d, **e;
```

```
char *f=p, *g=p;  
char **m=&p, **n=&p;
```

C's syntax for declaring multiple pointer variables on one line

`char* b, c;` does not work

C's syntax for declaring and initializing multiple pointer variables on one line

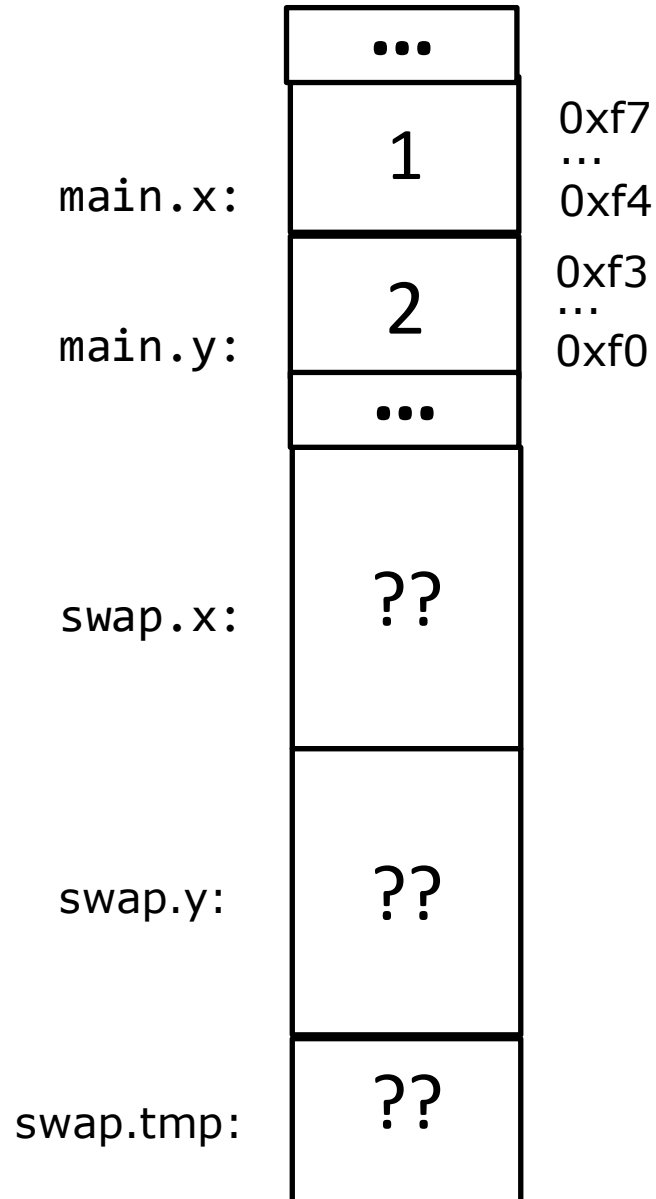
Pass pointers to function

```
void swap(int* x, int* y)
```

```
{  
  → int tmp = *x;  
    *x = *y;  
    *y = tmp;  
}
```

```
int main()  
{  
    int x = 1;  
    int y = 2;  
    swap(&x, &y);  
  
    printf("x:%d, y:%d", x, y);  
}
```

Size and value of x, y, tmp
in swap upon function entrance?

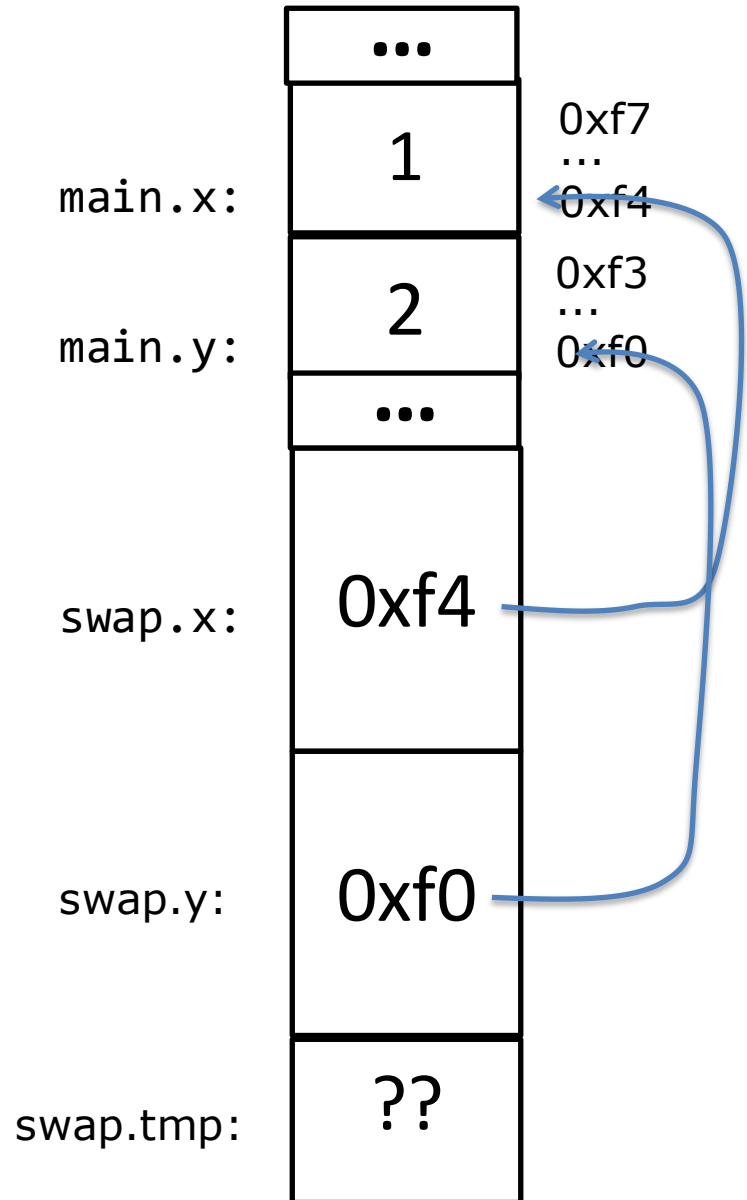


Pass pointers to function

```
void swap(int* x, int* y)
{
    int tmp = *x;
    *x = *y;
    *y = tmp;
}

int main()
{
    int x = 1;
    int y = 2;
    swap(&x, &y);

    printf("x:%d, y:%d", x, y);
}
```

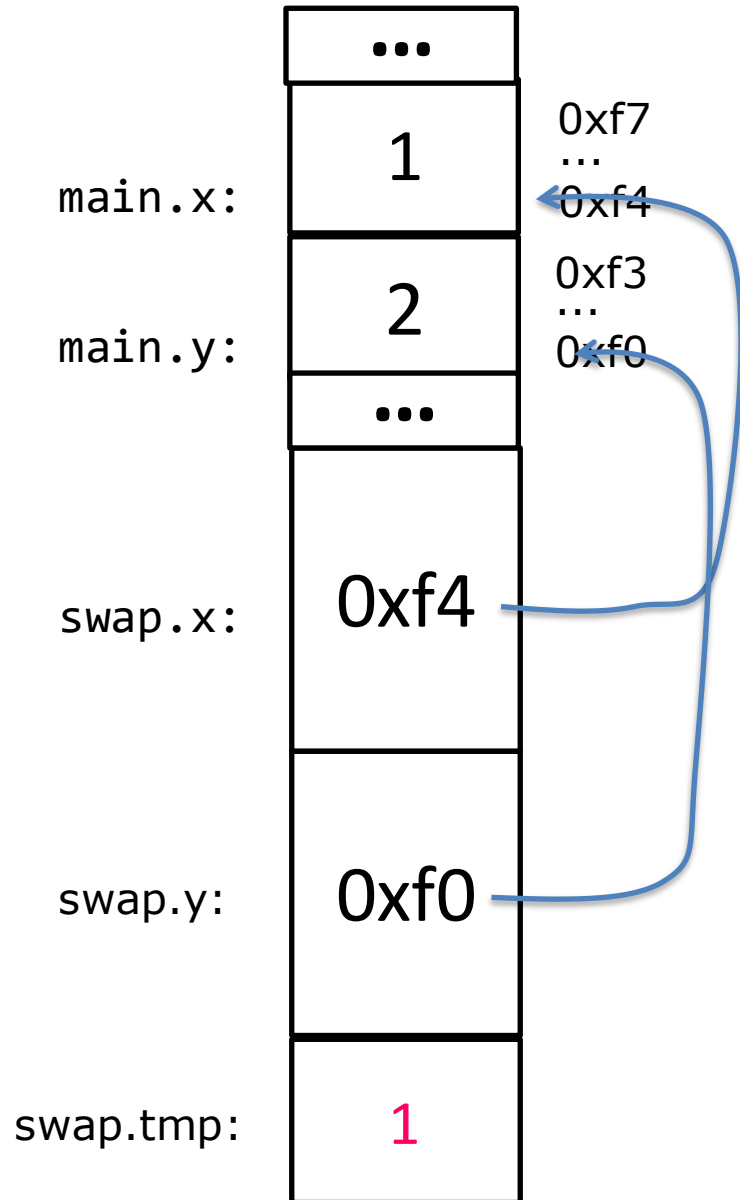


Pass pointers to function

```
void swap(int* x, int* y)
{
    int tmp = *x;
    *x = *y;
    *y = tmp;
}

int main()
{
    int x = 1;
    int y = 2;
    swap(&x, &y);

    printf("x:%d, y:%d", x, y);
}
```

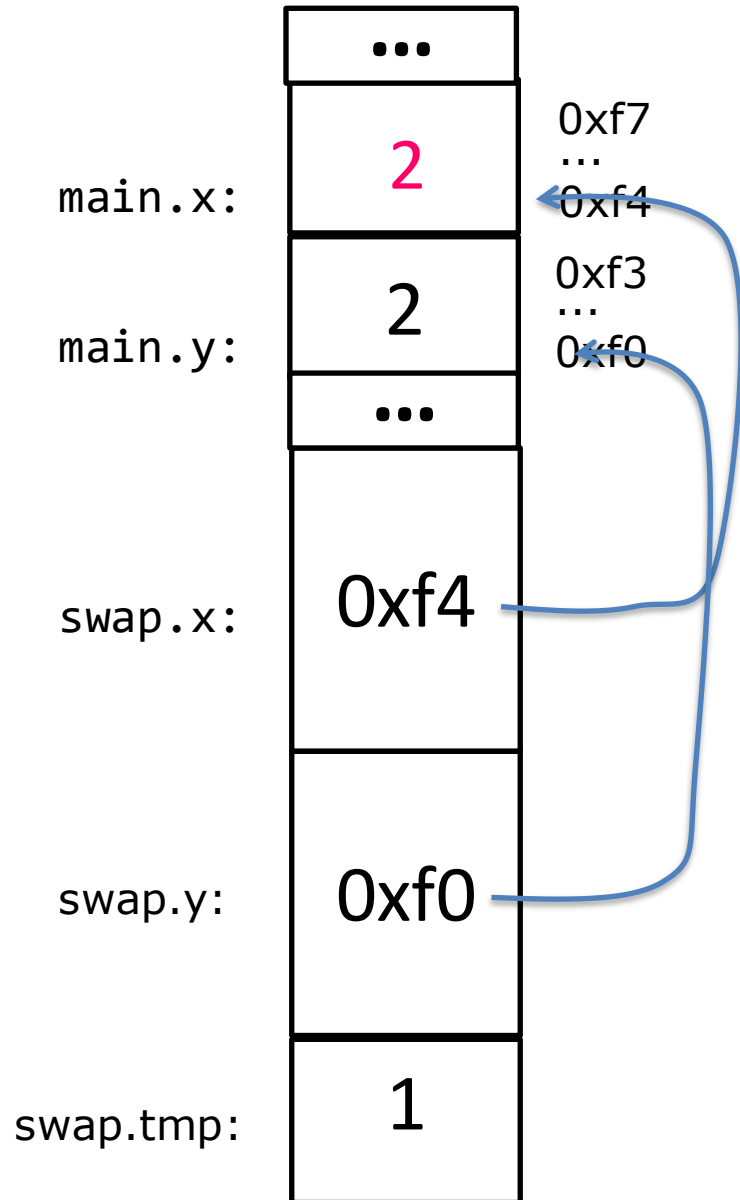


Pass pointers to function

```
void swap(int* x, int* y)
{
    int tmp = *x;
    *x = *y;
    *y = tmp;
}

int main()
{
    int x = 1;
    int y = 2;
    swap(&x, &y);

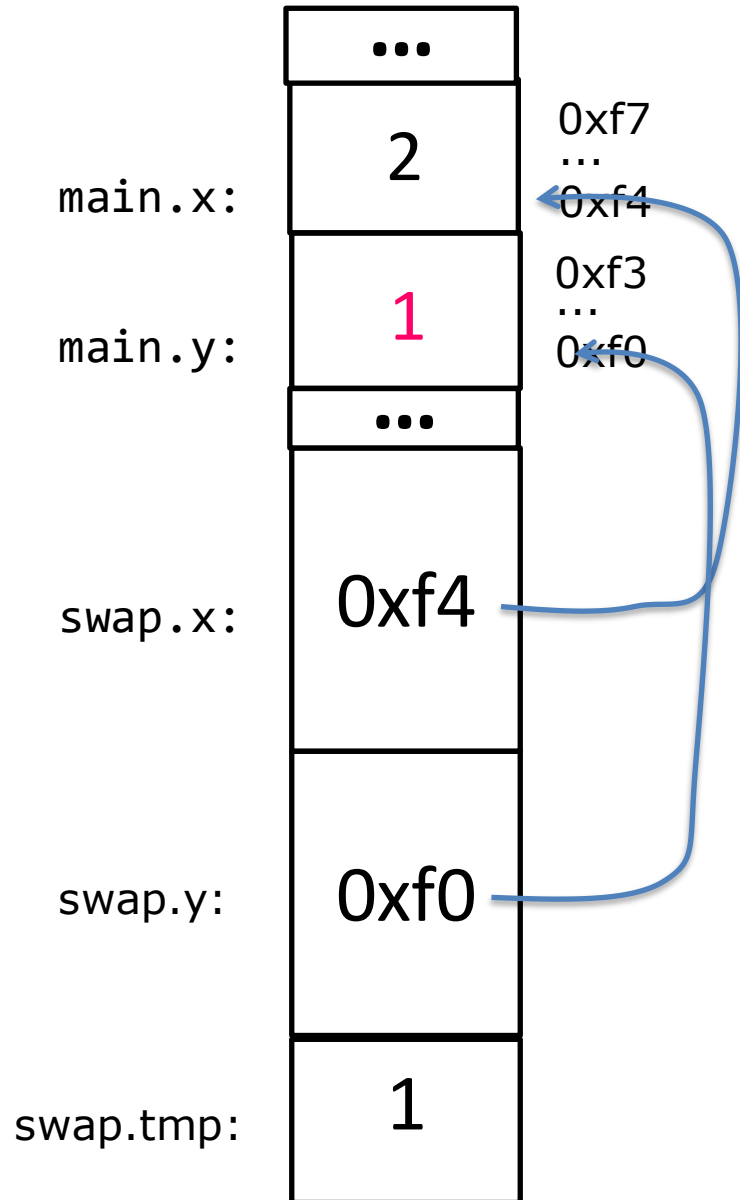
    printf("x:%d, y:%d", x, y);
}
```



Pass pointers to function

```
void swap(int* x, int* y)
{
    int tmp = *x;
    *x = *y;
    *y = tmp;
}
→
int main()
{
    int x = 1;
    int y = 2;
    swap(&x, &y);

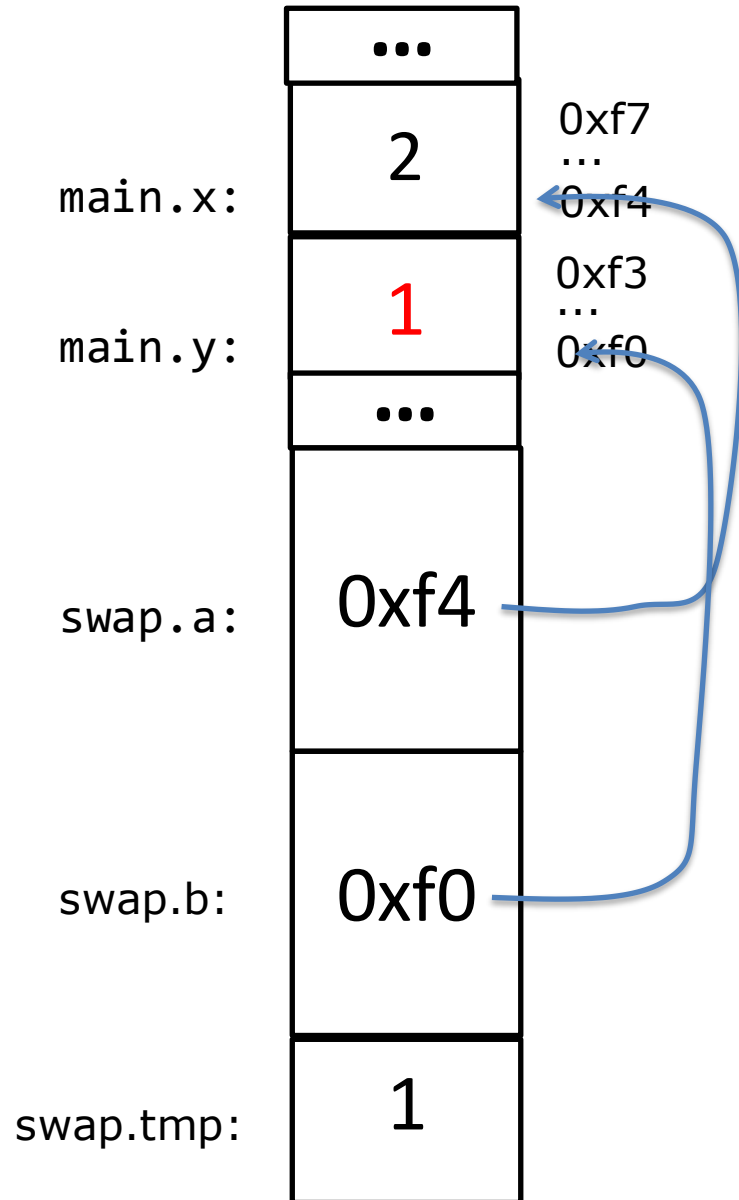
    printf("x:%d, y:%d", x, y);
}
```



Pass pointers to function

```
void swap(int* a, int* b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
}

int main()
{
    int x = 1;
    int y = 2;
    swap(&x, &y);
    printf("x:%d, y:%d", x, y);
}
```



Arrays

Array is a collection of contiguous objects with the same type

Array

a[3]:	0	0x11c
a[2]:	3	0x118
a[1]:	2	0x114
a[0]:	1	0x110
		...
		...
		...
		...

```
int a[4] = {1, 2, 3, 4};
```

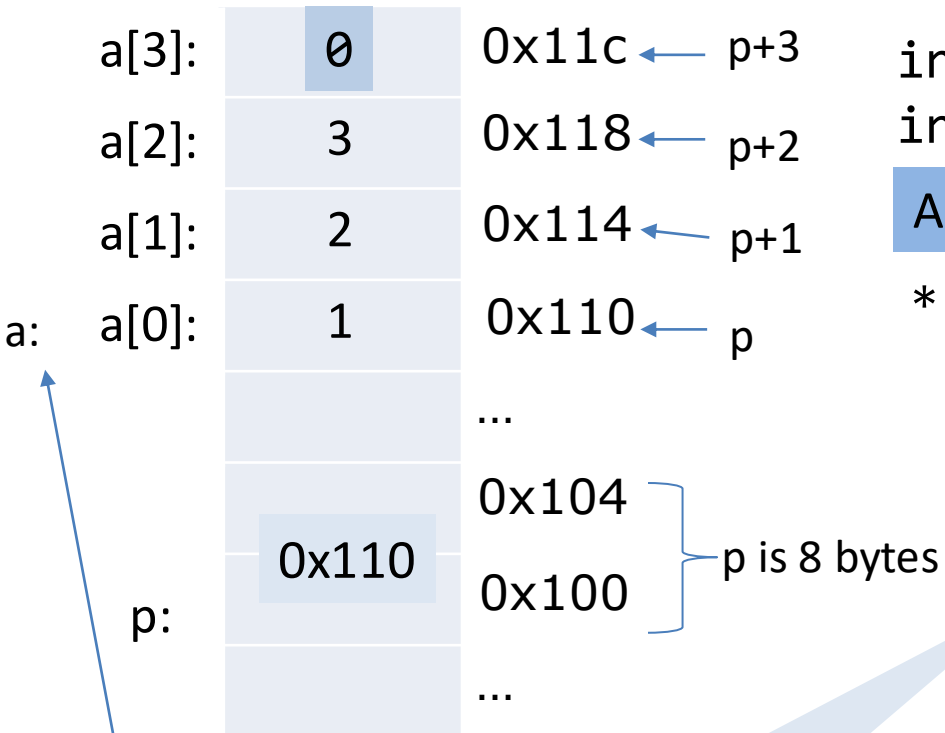
Access method-1: use index

```
a[3] = 0;
```



There's no meta-data (e.g. length) associated/stored with the array

Array access using pointer



```
int a[4] = {1, 2, 3, 4};  
int *p = a;
```

Access Method-2: use pointer (arithmetic)

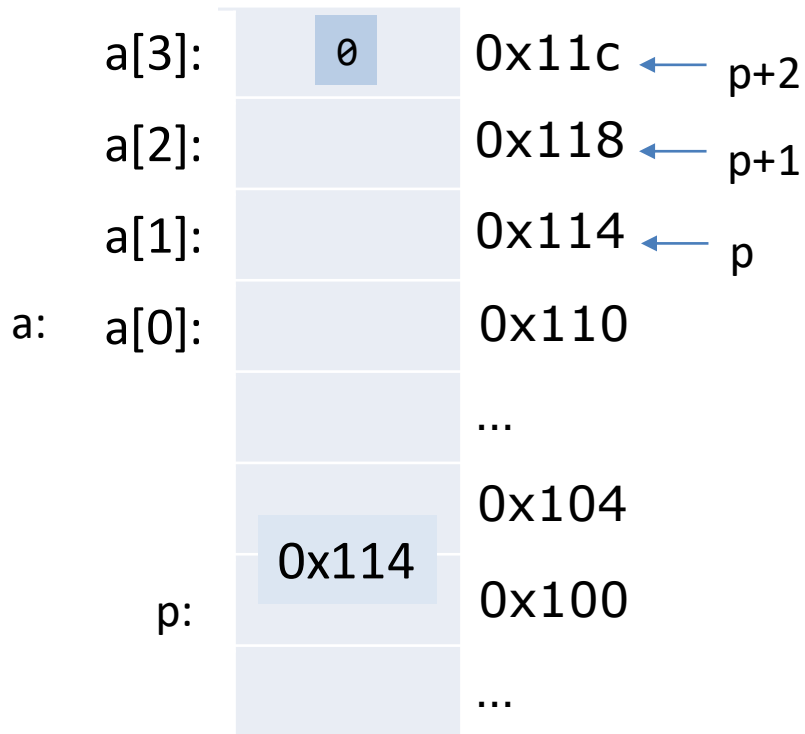
```
*(p+3)=0;
```

- `p+i` points to the i-th element after the one pointed to by `p`
i.e. `p+i` is calculated as: `p's value + sizeof(*p) * i`
`*(p+i)` is syntactically equivalent to `p[i]`
- `p-i` points to the i-th element before the one pointed to by `p`

Built-in function `sizeof` returns size (in bytes) of a given type or expression

`a` (array name) is aliased to be the memory address of the first element.
`a` is effectively a constant, not a variable, cannot be changed

Array access using pointer



`&a[i]` is syntactically equivalent to:

`a+i`

```
int a[4];
int *p = &a[1];
*(p+2)=0;
```

`*(p+i)` is syntactically equivalent to:

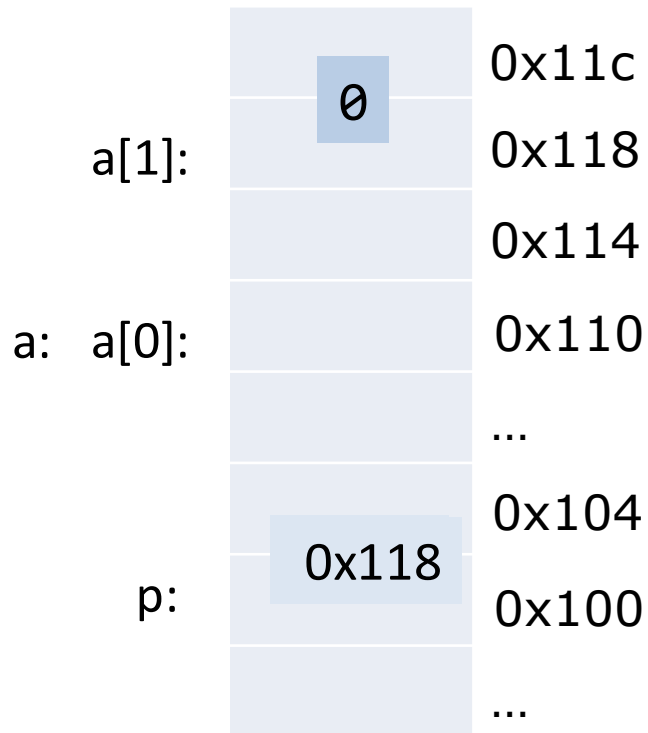
`p[i]`

Array access using pointer

a[3]:	4	0x11c
a[2]:	0	0x118
a[1]:	2	0x114
a: a[0]:	1	0x110
		...
		0x104
p:	0x11c	0x100
		...

```
int a[4] = {1, 2, 3, 4};  
int *p = &a[3];  
p--;  
*p=0;
```

Array access using pointer



```
char *a[2];
```

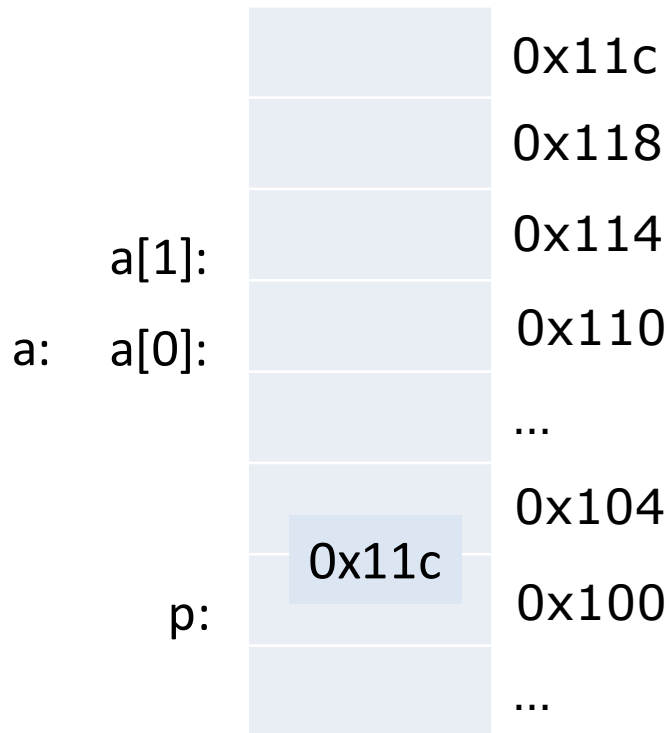
```
char** p = &a[0];
```

```
p++;
```

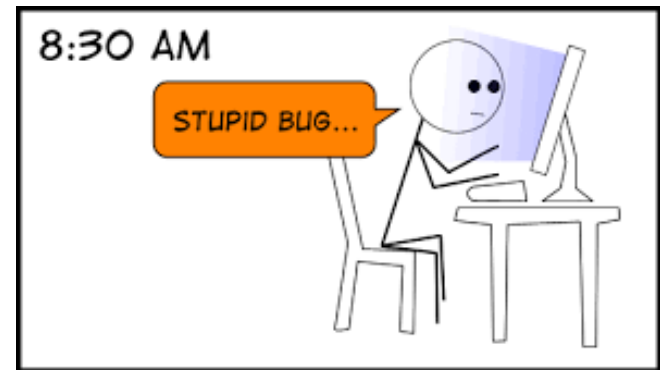
```
*p=NULL;
```

Equivalent to:
p = a

Out-of-bound access results in (potentially silent) memory error



```
int a[2];  
int *p = a;  
p += 3;  
*p=0;
```



Pass array to function via pointer

```
// multiply every array element by 2
void multiply2(int *a) {
    for (int i = 0; i < ???; i++) {
        a[i] *= 2;
    }
}
```

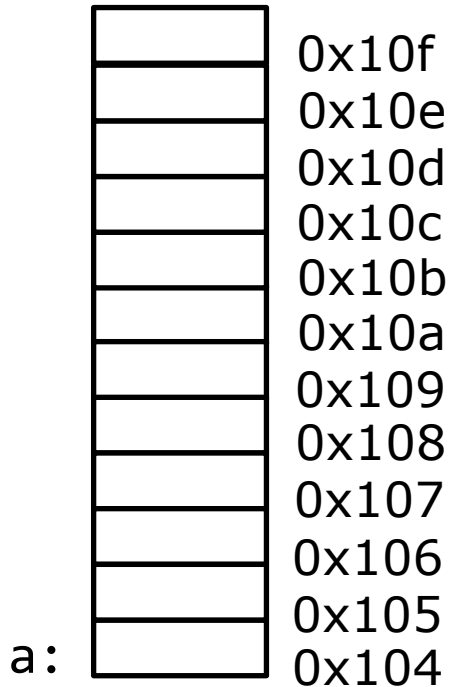
```
int main() {
    int a[2] = {1, 2};
    multiply2(a);
    for (int i = 0; i < 2; i++) {
        printf("a[%d]=%d", i, a[i]);
    }
}
```

Pass array to function via pointer

```
// multiply every array element by 2
void multiply2(int *a, int n) {
    for (int i = 0; i < n; i++) {
        a[i] *= 2; // (*(a+i)) *= 2;
    }
}
```

```
int main() {
    int a[2] = {1, 2};
    multiply2(a, 2);
    for (int i = 0; i < 2; i++) {
        printf("a[%d]=%d", i, a[i]);
    }
}
```

Pointer casting



```
int a = 0x12345678;  
char *cp = (char *)&a;  
  
// What is *cp?  
cp++;  
// What is *cp?
```

Assume 64-bit small endian machine

Another example use of pointer casting



```
unsigned char extract_float_exp_field(float f)
{
    unsigned int i = *(unsigned int *)&f;

}
```

Summary

- Pointers are memory addresses
 - `p = &x;` (p has address of variable x)
 - `*p ...` (refers to the variable pointed to by p)
- Arrays:
 - No array meta-data associated/stored. No bound checking
 - equivalence of pointer arithmetic and array access
 - `p+i` same as `&p[i]`
 - `*(p+i)` same as `p[i]`
 - Value of `p+i` is computed as `p+sizeof(*p)*I`
- Pass pointers to functions
- Pointer casting